Reproduction of "Macroscopic Quantum Superpositions via Dynamics in a Wide Double-Well Potential"

Roda-Llordes *et al.*, Phys. Rev. Lett. **132**, 023601 (2024)

Ilia Polishchuk · code and data: `https://github.com/ilpolishc/doublewell` · November 2026

What was done. I wrote, from scratch in Python/NumPy, a simulation of the single-particle protocol of the paper and reproduced its two results figures for the L parameter set of Table I ($X_s/D = 10^{-1}$, $\omega/\Omega = 10^{-2}$, $D/X_\Omega = 10^4$, $\bar{n} = 0$), with the XL/XXL curves where the paper shows them. The pieces are: (i) the classical orbit $u'' = u - u^3$, which defines a co-moving frame; (ii) a split-step Fourier propagator for the residual state in that frame ($n = 2^{17}$ points, half-width $4500\,X_\Omega$), with Wigner functions by FFT of the autocorrelation; (iii) the Gaussian squeezing $\mathcal{S}(t)$ from the quadratic covariance dynamics, as in the paper's endnote [41]; (iv) decoherence by two independent routes—stochastic momentum kicks on a 256-trajectory ensemble ("A") and the Gaussian-blur approximation with the width computed from the Green's function of the linearised dynamics ("B"). The constant-angle blur and the Route-B Green's-function blur are computed independently and agree to $< 1\%$ at L. (v) the robustness panel from full propagation of displaced initial states, re-weighted to each σ_s and $\bar{n}$; (vi) the two-particle relative coordinate $P(\tilde{r})$ as the self-convolution of the single-particle pattern (Supplemental Material); (vii) theanalytical treatment the paper's Fig. 3(b–d) lines come from (Ref. [37], Riera-Campeny et al., Quantum **8**, 1393): the constant-angle approximation, giving a closed-form Airy Wigner function at t_m from κ_3, κ_4, $S(t_\mathrm{m})$ and ϕ, with decoherence as a Gaussian blur of width $\sigma_\mathrm{b}^2 = 4(\Gamma/\omega)\int \eta^2 d\tau$. Visibility is the paper's own $V = (P_\mathrm{a} - P_\mathrm{b})/(P_\mathrm{a} + P_\mathrm{b})$ at the first minimum and adjacent maximum. Every figure was convergence-tested (box doubled, step halved), and all conventions and every mismatch found on the way are logged in the repository (`notes/`).

Paper, Fig. 2 | **This work**

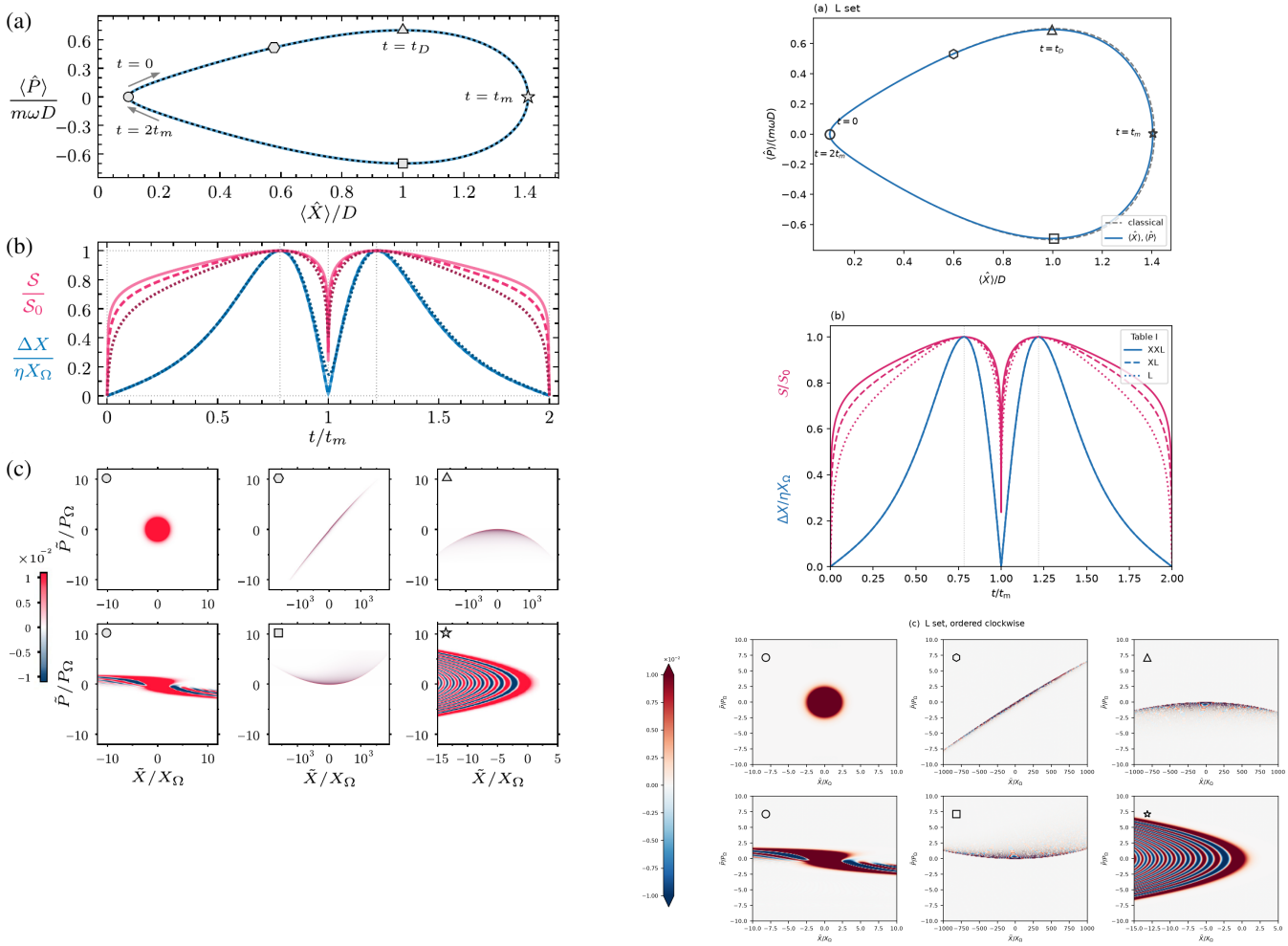


Figure 1: Coherent dynamics. (a) First moments $\langle\hat{X}\rangle, \langle\hat{P}\rangle$ over the classical orbit (dashed); the loop closes at $2t_\mathrm{m}$ and the far turning point sits at $\langle\hat{X}\rangle/D = \sqrt{2}$. (b) Normalised squeezing and position spread for XXL/XL/L; ΔX peaks at ηX_Ω at t_D and recompresses to $O(X_\Omega)$ at t_m. (c) Wigner function of the residual state at the six marked times (L set), with the same axes as the paper: squeezed line → bent "boomerang" → cubic-phase state with negativity fringes at t_m. Paper panels reproduced from Roda-Llordes *et al.*, Phys. Rev. Lett. **132**, 023601 (2024) (© APS), for comparison only.

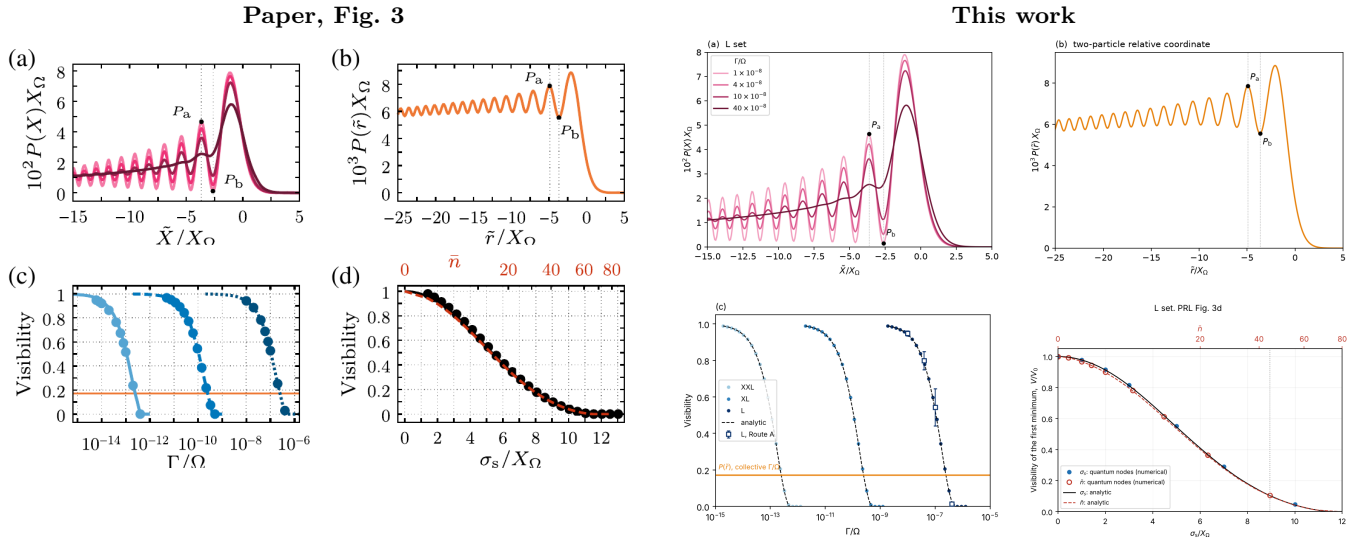


Figure 2: Decoherence and robustness. (a) $P(X, t_{\mathrm{m}})$ for $\Gamma/\Omega = \{1, 4, 10, 40\} \times 10^{-8}$ (L set), numerical pattern with the Gaussian-blur decoherence model (checked against a 256-trajectory stochastic-kick ensemble). (b) Relative-coordinate distribution $P(\tilde{r}, t_{\mathrm{m}})$ of two mirror-symmetric particles. (c) Visibility of the first minimum versus Γ/Ω for XXL/XL/L: dots numerical, dashed lines the analytical constant-angle result (as in the paper, where the lines are analytical); squares: stochastic-kick ensemble with its error. Horizontal line: visibility of $P(\tilde{r})$, immune to collective decoherence. (d) Visibility versus initial position imprecision σ_s and, on the top axis, initial occupation $\bar{n}$ (mapped through $\sigma = \sqrt{2\bar{n}}\, X_\Omega$): dots from full propagation of displaced states, lines analytical.

Table 1: Quantitative comparison. "Paper" values marked † were read off the published figure by digitising the axes (uncertainty $\approx \pm 0.03$ in V); all others are stated in the text or Table I. "Numerical" = split-step propagation (Route B blur unless marked A = stochastic kicks); "analytical" = my implementation of the constant-angle treatment of Ref. [37]. All from the data files in the repository.

Quantity	Paper	This work, numerical	This work, analytical	Comment
$\sqrt{2}\,\eta$; $\mathcal{S}_0$ (XXL/XL/L)	$10^5, 10^4, 10^3$; 97, 77, 57 dB	96.99, 76.99, 56.99 dB	–	exact, Eq. (1)
Orbit half-period ωt_{m}	$\ln(4\sqrt{2}D/X_s) = 4.036$	4.047 (exact quadrature)	–	+0.28%, asymptotic
$\Delta X(t_D)/X_\Omega$	$\eta = 707$	707 ($t_D/t_{\mathrm{m}} = 0.782$)	$\eta_\star = 707.1$	match, Fig. 2b
Fringe spacing X_f/X_Ω (L)	≈ 2.5	2.472 (2.506 on finer grid)	2.472	−1%
Coherent visibility V_0	1	0.9976	0.9979	match
Fig. 3c (L): V/V_0, $\Gamma/\Omega = 4 \times 10^{-8}$	0.78†	0.783 (A: 0.80)	0.783	match
at 1×10^{-7}	≈ 0.53†	0.528 (A: 0.54)	0.528	match
at 4×10^{-7}	0.03†	0.030 (A: 0.01)	0.029	match
Blur width σ/X_Ω, same four Γ	–	B: 0.115, 0.230, 0.363, 0.727	0.115, 0.230, 0.363, 0.727	A/B 1.00–1.16
Fig. 3b: peak of $10^3 P(\tilde{r}) X_\Omega$	≈ 8.7 at ≈ -2†	8.86 at $\tilde{r} = -2.06$	8.88 at −2.06	match
Fig. 3c: visibility of $P(\tilde{r})$	0.172†	0.172	0.172	match; flat in Γ
Fig. 3d: V/V_0 at $\sigma_s = 5, 7, 10\, X_\Omega$	0.53, 0.27, 0.04†	0.550, 0.289, 0.046	0.546, 0.285, 0.042	within ±0.02
Fig. 3d: V/V_0 at $\bar{n} = 40$	0.10†	0.105	0.101	match
Timing error $\sigma_t = 10^{-2}\omega^{-1}$	"visible pattern"	0.997 (0.81 at $10\sigma_t$)	–	confirmed
Decoherence scales as $\Gamma\eta^2/\omega$	stated in text	$\propto \eta^2/(\omega/\Omega)$ to 5 s.f.	same	confirmed (L, XL, M)
Wigner negativity	Fig. 2c, 3a	reproduced	Airy form	qualitative

Notes, limits and open points. The lines in the paper's panels 3b–d are analytical (its Ref. [37]). Implementing it allows me to compare my numerics and my analytical curves, they agree to ≤ 0.005 in V/V_0 in every panel of Fig. 2 (lines are drawn only in (c) and (d), where the paper has them; in (a) and (b) the two are indistinguishable), and the analytical blur width coincides with the exact linear-response (Route B) width to three figures at L; the only visible departure is the stochastic-kick ensemble at $\Gamma/\Omega = 4 \times 10^{-7}$,0.014 vs 0.029, inside its ensemble error. The remaining $\lesssim 0.02$ offsets against the published curves are at the level of reading the figure. In Fig. 3c the numerical XL and XXL points are the L blur curve mapped through the Green's-function integral computed for each set, whereas the analytical XL/XXL lines are computed directly for those sets ($\kappa_3 = 1000.0$ for all three, $X_f = 2.472$); in Fig. 2b all three sets come from the quadratic covariance dynamics. Only the L set was propagated as a full wavefunction. The position and momentum quadratures of the thermal mixture were propagated as separate node sets and composed multiplicatively. The momentum quadrature costs only $\approx 2\%$ ($V_{\mathrm{mom}} = 0.978$ at $\bar{n} = 40$).